SG No. 73 D_{2h}^{27} *Ibca* [orthorhombic]

* plus set: $+[0,\,0,\,0]$, $+\left[\frac{1}{2},\,\frac{1}{2},\,\frac{1}{2}\right]$
* Wyckoff site: `8a`, site symmetry: `-1`

Table 1: Wyckoff bond: `8a@8a`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X,\,Y,\,Z]$	$[0,\,0,\,0]$	`[1,-5]`
2	$[-X,\,-Y,\,Z]$	$\left[\frac{1}{2},\,0,\,\frac{1}{2}\right]$	`[2,-6]`
3	$[-X,\,Y,\,-Z]$	$\left[0,\,\frac{1}{2},\,\frac{1}{2}\right]$	`[3,-7]`
4	$[X,\,-Y,\,-Z]$	$\left[\frac{1}{2},\,\frac{1}{2},\,0\right]$	`[4,-8]`

* Wyckoff site: `8b`, site symmetry: `-1`

Table 2: Wyckoff bond: `8a@8b`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X,\,Y,\,Z]$	$\left[\frac{1}{4},\,\frac{1}{4},\,\frac{1}{4}\right]$	`[1,-5]`
2	$[-X,\,-Y,\,Z]$	$\left[\frac{1}{4},\,\frac{3}{4},\,\frac{3}{4}\right]$	`[2,-6]`
3	$[-X,\,Y,\,-Z]$	$\left[\frac{3}{4},\,\frac{3}{4},\,\frac{1}{4}\right]$	`[3,-7]`
4	$[X,\,-Y,\,-Z]$	$\left[\frac{3}{4},\,\frac{1}{4},\,\frac{3}{4}\right]$	`[4,-8]`

* Wyckoff site: `8c`, site symmetry: `2..`

Table 3: Wyckoff bond: `8a@8c`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[0,\,Y,\,Z]$	$\left[x,\,0,\,\frac{1}{4}\right]$	`[1,-4]`
2	$[0,\,-Y,\,Z]$	$\left[\frac{1}{2}-x,\,0,\,\frac{3}{4}\right]$	`[2,-3]`
3	$[0,\,-Y,\,-Z]$	$\left[-x,\,0,\,\frac{3}{4}\right]$	`[5,-8]`
4	$[0,\,Y,\,-Z]$	$\left[x+\frac{1}{2},\,0,\,\frac{1}{4}\right]$	`[6,-7]`

Table 4: Wyckoff bond: `8b@8c`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X,\,0,\,0]$	$\left[x,\,0,\,\frac{1}{4}\right]$	`[1,4]`
2	$[-X,\,0,\,0]$	$\left[\frac{1}{2}-x,\,0,\,\frac{3}{4}\right]$	`[2,3]`
3	$[-X,\,0,\,0]$	$\left[-x,\,0,\,\frac{3}{4}\right]$	`[5,8]`
4	$[X,\,0,\,0]$	$\left[x+\frac{1}{2},\,0,\,\frac{1}{4}\right]$	`[6,7]`

Table 5: Wyckoff bond: `16c@8c`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\,Y,\,Z]$	$\left[x,\,0,\,\frac{1}{4}\right]$	`[1]`
2	$[-X,\,-Y,\,Z]$	$\left[\frac{1}{2}-x,\,0,\,\frac{3}{4}\right]$	`[2]`
3	$[-X,\,Y,\,-Z]$	$\left[\frac{1}{2}-x,\,0,\,\frac{3}{4}\right]$	`[3]`
4	$[X,\,-Y,\,-Z]$	$\left[x,\,0,\,\frac{1}{4}\right]$	`[4]`

continued ...

Table 5

No.	vector	center	mapping
5	$[-X, -Y, -Z]$	$\left[-x, 0, \frac{3}{4}\right]$	`[5]`
6	$[X, Y, -Z]$	$\left[x+\frac{1}{2}, 0, \frac{1}{4}\right]$	`[6]`
7	$[X, -Y, Z]$	$\left[x+\frac{1}{2}, 0, \frac{1}{4}\right]$	`[7]`
8	$[-X, Y, Z]$	$\left[-x, 0, \frac{3}{4}\right]$	`[8]`

* Wyckoff site: `8d`, site symmetry: `.2.`

Table 6: Wyckoff bond: `8a@8d`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[\frac{1}{4}, y, 0\right]$	`[1,-3]`
2	$[-X, 0, Z]$	$\left[\frac{1}{4}, -y, \frac{1}{2}\right]$	`[2,-4]`
3	$[-X, 0, -Z]$	$\left[\frac{3}{4}, -y, 0\right]$	`[5,-7]`
4	$[X, 0, -Z]$	$\left[\frac{3}{4}, y, \frac{1}{2}\right]$	`[6,-8]`

Table 7: Wyckoff bond: `8b@8d`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0, Y, 0]$	$\left[\frac{1}{4}, y, 0\right]$	`[1,3]`
2	$[0, -Y, 0]$	$\left[\frac{1}{4}, -y, \frac{1}{2}\right]$	`[2,4]`
3	$[0, -Y, 0]$	$\left[\frac{3}{4}, -y, 0\right]$	`[5,7]`
4	$[0, Y, 0]$	$\left[\frac{3}{4}, y, \frac{1}{2}\right]$	`[6,8]`

Table 8: Wyckoff bond: `16c@8d`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[\frac{1}{4}, y, 0\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{4}, -y, \frac{1}{2}\right]$	`[2]`
3	$[-X, Y, -Z]$	$\left[\frac{1}{4}, y, 0\right]$	`[3]`
4	$[X, -Y, -Z]$	$\left[\frac{1}{4}, -y, \frac{1}{2}\right]$	`[4]`
5	$[-X, -Y, -Z]$	$\left[\frac{3}{4}, -y, 0\right]$	`[5]`
6	$[X, Y, -Z]$	$\left[\frac{3}{4}, y, \frac{1}{2}\right]$	`[6]`
7	$[X, -Y, Z]$	$\left[\frac{3}{4}, -y, 0\right]$	`[7]`
8	$[-X, Y, Z]$	$\left[\frac{3}{4}, y, \frac{1}{2}\right]$	`[8]`

* Wyckoff site: `8e`, site symmetry: `..2`

Table 9: Wyckoff bond: `8a@8e`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{4}, z\right]$	`[1,-2]`
2	$[-X, Y, 0]$	$\left[0, \frac{3}{4}, \frac{1}{2}-z\right]$	`[3,-4]`
3	$[-X, -Y, 0]$	$\left[0, \frac{3}{4}, -z\right]$	`[5,-6]`
4	$[X, -Y, 0]$	$\left[0, \frac{1}{4}, z+\frac{1}{2}\right]$	`[7,-8]`

Table 10: Wyckoff bond: `8b@8e`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[0,\ \frac{1}{4},\ z\right]$	`[1,2]`
2	$[0,\ 0,\ -Z]$	$\left[0,\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[3,4]`
3	$[0,\ 0,\ -Z]$	$\left[0,\ \frac{3}{4},\ -z\right]$	`[5,6]`
4	$[0,\ 0,\ Z]$	$\left[0,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[7,8]`

Table 11: Wyckoff bond: `16c@8e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$\left[0,\ \frac{1}{4},\ z\right]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[0,\ \frac{1}{4},\ z\right]$	`[2]`
3	$[-X,\ Y,\ -Z]$	$\left[0,\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[3]`
4	$[X,\ -Y,\ -Z]$	$\left[0,\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[4]`
5	$[-X,\ -Y,\ -Z]$	$\left[0,\ \frac{3}{4},\ -z\right]$	`[5]`
6	$[X,\ Y,\ -Z]$	$\left[0,\ \frac{3}{4},\ -z\right]$	`[6]`
7	$[X,\ -Y,\ Z]$	$\left[0,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[7]`
8	$[-X,\ Y,\ Z]$	$\left[0,\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[8]`

* Wyckoff site: `16f`, site symmetry: `1`

Table 12: Wyckoff bond: `16a@16f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X,\ Y,\ Z]$	$[x,\ y,\ z]$	`[1]`
2	$[-X,\ -Y,\ Z]$	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	`[2]`
3	$[-X,\ Y,\ -Z]$	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[3]`
4	$[X,\ -Y,\ -Z]$	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	`[4]`
5	$[-X,\ -Y,\ -Z]$	$[-x,\ -y,\ -z]$	`[5]`
6	$[X,\ Y,\ -Z]$	$\left[x+\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$	`[6]`
7	$[X,\ -Y,\ Z]$	$\left[x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[7]`
8	$[-X,\ Y,\ Z]$	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	`[8]`